Problem 1: PPN and Gravitational Red Shift

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April 4, 2026

1 Problem

Question: Is the experiment of Vessot & Levine sensitive enough to say anything about the parameters β and γ? Is the third-order doppler effect important in analyzing the experiment?[1]

1.1 Introduction

This problem is an exercise in applying the Parameterized post-Newtonian (PPN) framework to an experimental background. Specifically, by expressing the gravitational frequency shift in terms of the PPN parameters, one can deduce whether experiments testing gravitational redshift, such as Vessot et al. (1980), can be used to test differing metric theories of gravity. Before presenting an explicit solution to the problem I will first sketch both the frequency shift as predicted by the Equivalence Principle (and a more explicit prediction using the Schwarzchild metric). We will then explore, briefly, the experimental design of Vessot et al. Then we will describe the PPN framework, insofar as it applies to this problem.

1.2 The Gravitational Redshift

The gravitational frequency shift is a consequence of the Equivalence Principle. If it is accepted that uniformly frame is entirely equivalent to a uniform

[1] Question 10.2. from J. B. Hartle and Cambridge University Press, Gravity: an introduction to Einstein's general relativity. Cambridge Cambridge University Press, 2021.

(or local) gravitational field, then it must be the case that the rates of emission and reception for spatially separated observers are different. That is to say that frequencies are shifted, preportional to the gravitational field strength. All theories of gravity that are reconciled with the Equivalence Principle—which is to say, all modern theories of gravity—will predict a gravitational frequency shift. The equivalence principle explicitly predicts that for a detector B and an emitter A the rates of emission and reception (i.e. frequencies ω_B and ω_A) are related by the difference in the gravitational field strengths[2]

$$\omega_B = \left(1 + \frac{\Phi_A - \Phi_B}{c^2}\right)\omega_A. \tag{1}$$

As we will see shortly, the Vessot and Levin experiment involves the frequency shift involving signals on the surface of the Earth and in orbit. So we can apply (1) to a situation where a signal is emitted from the surface of the Earth and measured by an observer far away. In this case the gravitational potential on the surface is

$$\Phi = -\frac{GM_\oplus}{R_\oplus} \tag{2}$$

where $M_\oplus$ and $R_\oplus$ is the mass and radius of the Earth respectively. The potential as $r \to \infty$ is zero. Thus we have

$$\omega_\infty = \left(1 - \frac{GM_\oplus}{R_\oplus c^2}\right)\omega_\oplus. \tag{3}$$

This problem can treated in General Relativity by approximating the gravitation field using the Schwarzchild geometry, produced by a spherically symmetric body. The Schwarzchild Metric, for the spacetime approximation, is given in geometrical units

$$ds^2 = -\left(1 - \frac{2M}{r}\right)dt^2 + \left(1 - \frac{2M}{r}\right)^{-1} dr^2 + r^2(d\theta^2 + \sin^2\theta d\phi^2). \tag{4}$$

The gravitational redshift, (3), can be recovered in a clever way. First we note that the Killing vector has temporal and spherical symmetry; the metric (4) is independent of the coordinates t and ϕ. For our purposes we will exploit the temporal symmetry and define a Killing vector

[2]See Hartle, 113-119

$$\xi^\alpha = (1, 0, 0, 0). \tag{5}$$

We then define the conserved scalar product of the four-velocity and this Killing vector along a geodesic

$$\boldsymbol{\xi} \cdot \boldsymbol{u} = \text{const.} \tag{6}$$

Next we set up on problem as following: an observer travels along a world line with four-velocity $\boldsymbol{u}_{obs}$ measures the frequency of light by measuring its energy. The energy an observer measures is related to the four-momentum of the light and their own four-velocity by

$$E = -\boldsymbol{p} \cdot \boldsymbol{u}_{obs}. \tag{7}$$

Furthermore we can relate the energy of a photon to its frequency using the Planck relation. This permits the following relation

$$\hbar\omega = -\boldsymbol{p} \cdot \boldsymbol{u}_{obs}. \tag{8}$$

We note that the spatial components of the observers four-velocity are all zero. Using the normalization condition of four-velocity, we have

$$g_{tt}[u^t_{obs}]^2 = -1. \tag{9}$$

Using the temporal metric element from (4) we can specify the four-velocity component

$$u^t_{obs} = \left(1 - \frac{2M}{r}\right)^{-\frac{1}{2}}. \tag{10}$$

Using the Killing vector (5) we can thus write the observers four-velocity as

$$\boldsymbol{u}_{obs} = \left(1 - \frac{2M}{r}\right)^{-\frac{1}{2}} \boldsymbol{\xi}. \tag{11}$$

The frequency of light as measured by an observer on the surface of the Earth is

$$\hbar\omega_{R_\oplus} = \left(1 - \frac{2M}{r}\right)^{-\frac{1}{2}} (\boldsymbol{p} \cdot \boldsymbol{\xi})\,|_{r=R_\oplus}. \tag{12}$$

Similarly, the frequency of light as measured by an observer far away as $r \to \infty$ is

$$\hbar\omega_\infty = \left(1 - \frac{2M}{r}\right)^{-\frac{1}{2}} |_{r\to\infty} \left(\boldsymbol{p}\cdot\boldsymbol{\xi}\right)|_{r\to\infty}. \tag{13}$$

The Schwarzchild coefficient reduces to one in the limit where $r \to \infty$. We finally note the conserved scalar product as defined by (6). This permits (13) to be substituted into (12) and we find

$$\frac{\omega_\infty}{\omega_{R_\oplus}} = \left(1 - \frac{2M}{r}\right)^{\frac{1}{2}}. \tag{14}$$

This result can be expanded in powers of $2M/R$ when that is small. The first order term recovers the redshift as prediced solely by the Equivalence Principle in (3).

1.3 The Experiment of Vessot et al.

The predictions of the gravitational frequency shift have been tested experimentally to quite amazing degrees of accuracy. It is worth noting that these experiments do not stricly test General Relativity. In fact, that is the entire point of this problem. All theories of gravity, that are in agreement with the Equivalence Principle, predict the frequency shift. Thus, experiments such as those described below are really tests of the Equivalence Principle; they do not, in and of themselves, discriminate between different forms of metric theories of gravity.

Vessot and Levin (1980)[3] was an experiment conducted to test the relativistic frequency shift as predicted by the Equivalence Principle. The experiment was conducted using frequency comparisons generated from hydrogen masers seperated by a gravitational potential. The ground system was located at Merrit Island, Florida and the other maser/transponder system was launched on the Scot Rocket, a 100-kg spin-stabilized spacecraft, with an altitude of 10 000 km. The experimental design is quite ingenious in the fact that it was able to eliminate a major source of error. There are two sources of relativistic frequency shift: the doppler shift, a consequence of the relative motion of the spacecraft and the ground system, and the gravitational redshift. The former can be expressed as a Taylor series, whose first and second order terms are of principle importance. The first order term is on the order

[3] R. F. Vessot et al., "Test of relativistic gravitation with a space-borne hydrogen maser," *Physical Review Letters*, vol. 45, no. 26, pp. 2081–2084, Dec. 1980. doi:10.1103/physrevlett.45.2081

$\frac{V}{c}$, where V is the relative velocity. In the experiment this would result in a shift on the order 10^{-5}. The second order doppler shift would thus be on the order 10^{-10}. On the other hand the gravitational redshift would be on the order 10^{-10} as well. It is clear that the first order doppler shift would dominate. Any experiment would need to isolate or remove the first order doppler effect in order to probe the gravitational phenomena.

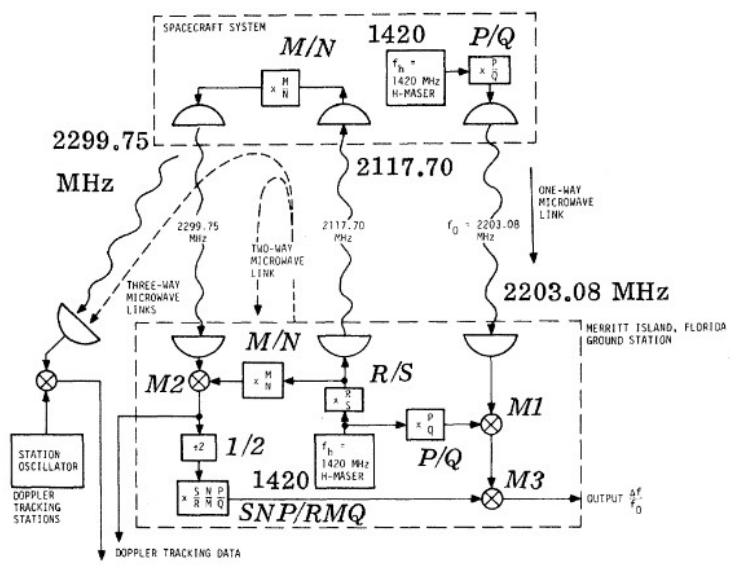

Figure 1: Schematic of the Vessot & Levine gravitational redshift experiment.

Figure 1 is a sketch of the Doppler cancellation and tracking system as used by Vessot et al. Two masers are used, with two-way maser-transponder-mixer system used to isolate the first order Doppler shift. The authors of the paper note that the first order Doppler effects "were removed by subtracting in mixer M3 one-half the two-way Doppler shift obtained from mixer M2 by comparing the separate two-way transponder links." The result is a measurement of the frequency shift whose effects include only that of second order Doppler effect and the gravitational redshift. The former can be isolated due to the fact that the velocities of the ground system and the probe (spacecraft) system are known.

1.4 Parameterized post-Newtonian Frameworks

Parameterized post-Newtonian (PPN) formalism is a framework in which different metric theories of gravity can be compared with each other, and be experimentally tested. Metric theories of gravity are those theories that posit that gravitation arises from curvature of spacetime. These theories include General Relativity, the most well known and tested, as well as others. The PPN framework is quite sophisticated and still used in more modern studies of theories of garvity. For example in Keeton and Petters (2006),[4] PPN was used to express gravitational lensing by compact objects in terms of the PPN parameters. The authors note that such a study presents "new [prospects] for testing theories of gravity within the PPN framework."[5] Further mock experiments have been proposed, by Losada et al., to constrain measures of PPN parameters.[6] The proposal suggests using measurements for objects orbiting Sagittarius A* in order to constraint the first-order PPN parameters. Application of this method can be used in the future for "providing a precision test of general relativity and its extensions."

Metric theories of gravity, for the most part, provide a spherically symmetric with the form

$$ds^2 = -A(r)(cdt)^2 + B(r)dr^2 + r^2(d\theta^2 + \sin^2\theta d\phi^2). \tag{15}$$

The functions $A(r)$ and $B(r)$ are specified by the first post-Newtonian corrections that are in agreement with the weak-field limit

$$A(r) = \left(1 - \frac{2GM}{c^2 r} + 2(\beta - \gamma)\left(\frac{GM}{c^2 r}\right)^2\right) \tag{16}$$

$$B(r) = 1 + 2\gamma\left(\frac{GM}{c^2 r}\right) \tag{17}$$

Differing metric theories of gravity will propose different values for the PPN parameters β and γ. General Relativity, for example, has values $\beta = 1$ and $\gamma = 1$. The utility of the PPN framework is material observations, such

[4] C. R. Keeton and A. O. Petters, "Formalism for testing theories of gravity using lensing by compact objects. II. probing post-post-newtonian metrics," *Physical Review D*, vol. 73, no. 4, Feb. 2006. doi:10.1103/physrevd.73.044024

[5] Keeton and Petters, 10

[6] [V. de Mora Losada, R. Della Monica, I. de Martino, and M. De Laurentis, "Future prospects for measuring 1PPN parameters using observations of S2 and S62 at the Galactic Center," Astronomy & Astrophysics, vol. 694, Feb. 2025. doi:10.1051/0004-6361/202452814

as the deflection of light by the Sun, or the precession of the perihelion of a planets orbit, or the Shapiro time delay, can be expressed in terms of the PPN parameters. Thus experiments can be used to determine the values of the PPN parameters; and compare them with the proposed values from a given metric theory of gravity.

2 Solution

We will use the PPN metric given in (15) and the function expansions in terms of the PPN parameters as given by (16) and (17). Using this metric we can repeat the the general gravitational redshift problem. Recall the structure: we take an emitter and observer at rest. In the case of Vessot & Levine, we are considering an emitter on the surface of the Earth, and an observer at rest some large R away. The observer has a four-velocity $\boldsymbol{u}_{obs}$. The energy measured by the observer is

$$E = -\boldsymbol{p} \cdot \boldsymbol{u}_{obs}. \tag{18}$$

We relate this scalar product to the frequency of the photon using Planck's relation

$$\hbar\omega = -\boldsymbol{p} \cdot \boldsymbol{u}_{obs}. \tag{19}$$

The spatial components of the observers four-velocity are zero; that is $u^i_{obs} = 0$. We can then use the normalization constraint on the four-velocity

$$\boldsymbol{u}_{obs} \cdot \boldsymbol{u}_{obs} = g_{\alpha\beta} u^\alpha_{obs} u^\beta_{obs} = -1. \tag{20}$$

With the vanishing spatial components we are left with

$$g_{tt}[u^t_{obs}]^2 = -1. \tag{21}$$

The metric element is given by (16)

$$-A(r)[u^t_{obs}]^2 = -1. \tag{22}$$

Explicitly then, the only non-vanishing part of the observers four-velocity is

$$u^t_{obs} = A(r)^{-\frac{1}{2}}. \tag{23}$$

We can also use the Killing vector ξ, as described above, to express the temporal component of the four-velocity as

$$u^{\alpha}_{obs} = A(r)^{-\frac{1}{2}}\xi^{\alpha}. \tag{24}$$

Explicity the four-velocity is

$$\boldsymbol{u}_{obs} = A(r)^{-\frac{1}{2}}\boldsymbol{\xi}. \tag{25}$$

From this point forward we can exploit the symmetry of the spacetime, exactly as it is done in the Schwarzchild instance. We use (12) with $A(r)$ replacing the Schwarzchild coefficient, and evaluate using $r = R_{\oplus}$, where $R_{\oplus}$ is the radius of the Earth,

$$\hbar\omega_{R_{\oplus}} = A(R)^{-\frac{1}{2}}\left(\boldsymbol{p}\cdot\boldsymbol{\xi}\right)|_{r=R_{\oplus}}. \tag{26}$$

We also consider the distance observer, where for our purposes we let $r \to \infty$

$$\hbar\omega_{\infty} = A(\infty)^{-\frac{1}{2}}\left(\boldsymbol{p}\cdot\boldsymbol{\xi}\right)|_{r\to\infty}. \tag{27}$$

$A(\infty)$ reduces to unity which simplifies the expression. The conserved quantity (6) can be used to relate the two frequencies. We use (27), simplified with the PPN function reducing to unity, and insert the result into (26). This yields

$$\hbar\omega_{R_{\oplus}} = A(R_{\oplus})^{-\frac{1}{2}}\hbar\omega_{\infty}, \tag{28}$$

and the gravitational red shift resolves to

$$\frac{\omega_{\infty}}{\omega_{R_{\oplus}}} = A(R_{\oplus})^{\frac{1}{2}}. \tag{29}$$

Consider the expansion of the function $A(R_{\oplus})$ in terms of the PPN parameters

$$A(R_{\oplus}) = 1 - \frac{2GM_{\oplus}}{c^2R_{\oplus}} + 2\left(\beta-\gamma\right)\left(\frac{2GM_{\oplus}}{c^2R_{\oplus}}\right)^2 \tag{30}$$

The linear term will be on the order

$$\frac{2GM_{\oplus}}{c^2R_{\oplus}} \approx 10^{-5}, \tag{31}$$

whereas the quadratic term will be on the order

$$\left(\frac{2GM_{\oplus}}{c^2R_{\oplus}}\right)^2 \approx 10^{-10}. \tag{32}$$

The quadratic corrections, which are tuned by the PPN parameters, will thus be silenced by the dominant linear term.

For the second part of the question, regarding the third-order Doppler effect, we note that the effect would be on the order

$$\left(\frac{V}{c}\right)^3 \approx 10^{-15}. \tag{33}$$

This differs 10^{-5} from the second-order Doppler effect. The Vessot et al. experiment however only offers an accuracy up to 10^{-4}. Therefore the third-order effects are not important in the experiment.

3 Conclusion

The solution sketched above highlights the fact that the Vessot et al. experiment cannot be used to discriminate or constrain PPN parameters. In other words, it cannot be used to test and contrast differing metric theories of gravity. As mentioned in the introduction, this is not entirely suprising. Vessot et al., and more generally experiments concerning the gravitational frequency shift are tests of the *Equivalence Principle*. Gravitational redshift is a consequence of the equivalence of uniformly accelerated frames of reference and frames of reference subsumed in a uniform gravitational field. So long as the metric theory agrees with the Equivalence Principle, or in an equivalence statement, so long as the theory is complaint with the first-order post-Newtonian weak-field limit, then the theory predicts gravitational redshit.

The PPN framework becomes useful when testing physical phenomena, predicted by metric theories of gravity, for which effects are dominated PPN terms. The deflection of light by the Sun, or other compact objects for instance, are dominated by parameters γ. It is in these studies where the differing metric theories of gravity can be tested. And, thus far, General Relativity remains king amongst them.